

Project Proposal

Project Name: A Luminous Library

Abstract: I read a lot of books...nearly 40 last year, so I wanted a way to classify them all. The best way I could think of doing this is to make a smart book tracker that will automatically track what I'm reading. I will be implementing a book tracker that will display the genre and progress of each book based on the state of the program. So, a person must press a button once to start the registration process. Then, a person can use 2 DIP Switches to choose between 16 different genres. Once a genre has been chosen, use a button to register that value and to switch to progress selection state. The progress selection is controlled by a potentiometer and displayed by both the LEDs and the display. Once the progress is chosen, the data validation feedback is through small chimes using a speaker and amplifier as well. Finally, the data will be tracked using an online database through Google's Web Dev program **Firebase**. There is also a gesture tracking photoresistor that will enter a different state when a hand is held over the photoresistor. The switches will be used to choose the genre once again and then press the button to get a special recommendation for that genre!

Project Specifications:

Usage	Component
Analog Input	Progress Bar Potentiometer
Analog Output	DAC for Chime
Digital Input	2 x DIP Switches
Digital Output	5 x LEDs
Voltage Regulation (9V - 5V)	LM2576 Switching Regulator
Audio Amplification/Speaker	LM386 + 8 ohm speaker
Display Progress and Genre	LCD Display (I2C)
Extra	Gesture Tracking Photoresistor + Registering Button

Milestone 1: Configure Breadboard and Test Each Component

- Make sure I can output the chime
- Make sure the switches can detect different states
- Test the 8-Ohm speaker and amplifier
- Test the Digital Input - Assign values for each genre
- Test the Digital Output - Learn PWM for each of the LEDs

Milestone 2: Programming the Circuit

- Implement working functionality with the LCD display
- Test to make sure the circuit functions with a 9V current-limiting source
- Possibly implement real-time control between the progress bar and the potentiometer

Milestone 3: Implement with Firebase

- Test the Pico 2W's Bluetooth / Wifi capabilities + implement tracking with Firebase